Estimation of Spatiotemporal Variability of Global Surface Ocean DIC Fields Using Ocean Color Remote Sensing Data

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***Abstract*— The estimation of dissolved inorganic carbon (DIC) in global surface ocean waters is crucial for understanding air-sea carbon dioxide (CO_2) flux rates, ocean acidification, and climate change. DIC magnitude and spatiotemporal variability are influenced by various physical and biogeochemical processes. Due to dynamic variations in ocean surface water, estimating DIC through in situ data alone is challenging. Ocean color remote sensing offers high spatial and temporal resolution data with extensive synoptic views. Over decades, multiple DIC approaches have emerged using in situ and satellite observations but are limited to specific regions due to improper model parameter selection and sparse in situ measurements. To address this, we propose a novel multiparametric regression (MPR) approach that relates DIC as a function of sea surface temperature (SST), sea surface salinity (SSS), and chlorophyll-a (Chl*a*) concentration. Utilizing in situ data from the Global Ocean Data Analysis Project (GLODAP), the trends of DIC with SST, SSS, and Chla were analyzed to develop MPR regression equations. The validation results indicated that the proposed regression approach accurately estimates DIC in global surface ocean waters. This approach offers benefits, such as DIC estimates at any spatiotemporal resolutions, easy implementation, and cost-effective alternatives to in situ measurements. Additionally, seasonal and interannual variations of global DIC fields were demonstrated through satellite oceanographic data, enhancing monitoring of ocean acidification and climate change scenarios.**

***Index Terms*— Carbon cycle, dissolved inorganic carbon (DIC), global ocean, multiparametric regression (MPR).**

I. Introduction

DISSOLVED inorganic carbon (DIC, μmol kg^{-1}) is a measure of the amount of carbon dioxide (CO_2) and other inorganic forms of carbon that are dissolved in ocean waters. Over the past century, anthropogenic activities (burning of fossil fuels, deforestation, and land use changes) [1] have led to a significant increase in atmospheric CO_2 concentrations, which in turn caused an increase in the amount of DIC in the global oceans. The increased concentration of DIC is one of the key drivers of ocean acidification, which is a major concern for marine ecosystems and global carbon cycling [1], [2], [3], [4], [5]. The estimation of the magnitude and spatiotemporal variability of surface ocean DIC plays a predominant role in a better understanding of air-sea CO_2 flux rates and changes in the biogeochemistry of the ocean [3]. The concentration of DIC fields is majorly influenced by a complex array of physical (physical mixing, advection, diffusion, stratification, ocean currents, and circulations), biological (photosynthesis and respiration), and chemical processes (precipitation and dissolution) [1], [4]. These processes are controlled by governing parameters, such as sea surface temperature (SST) [6], [7], [8], sea surface salinity (SSS) [6], [7], chlorophyll-a concentration (Chla) [6], [9], dissolved oxygen (DO) [10], apparent oxygen utilization (AOU) [7], silicate (Si) [10], phosphate (P) [7], nitrate (Ni) [11], latitude [7], and depth [7].

Understanding the relative importance of these processes and their corresponding governing parameters is crucial for predicting how DIC concentrations may change in the future and assessing the impacts of these changes on marine ecosystems and the global climate change. In situ observations (ships/buoys/moorings) are one way to measure the spatiotemporal variations of DIC in the global oceans. However, in situ observations have several challenges, such as being time-consuming, expensive, and difficult to access in remote regions or in harsh weather conditions [1], [12]. To overcome these challenges, researchers have developed remote sensing techniques that use satellite data to estimate DIC concentrations in the ocean [6], [7], [8], [9]. These techniques provide a more comprehensive picture of DIC variability than in situ data alone, as they can cover large areas of the ocean and provide data over long time periods.

Based on the greater advantage of satellite oceanographic data, multiple in situ and satellite-based DIC algorithms have been developed over various major oceans, such as the Indian Ocean [7], Pacific Ocean [9], Atlantic Ocean, Southern Ocean [10], and Arctic Ocean [6]. Wallace [13] first confirmed the concept of multiple linear regression (MLR) approach by effectively predicting DIC in the North Atlantic Ocean through

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the use of SST, SSS, Si, and AOU. Similarly, Bates et al. [7] investigated the spatiotemporal fluctuations of surface ocean DIC fields in the Indian Ocean by correlating them with SST, SSS, AOU, Ni, P, latitude, and depth. In the Southern Ocean, McNeil et al. [10] developed a predictive empirical relationship of DIC with hydrographic properties (SST, SSS, DO, N, and Si) obtained from the World Ocean Atlas. Furthermore, Arrigo et al. [6] utilized an MLR model that incorporated SST, SSS, and Chla to determine DIC concentrations in the Arctic Ocean. While regional DIC models can be useful for understanding DIC variability within a specific region, they may not be adequate for producing global DIC fields due to their limitations in capturing the full range of spatiotemporal variability across different latitudes, longitudes, and ocean basins. Although the estimation of global-scale DIC fields is possible by combining various regional models, which poses a significant challenge due to the complexity of the multiple relationships involved at different spatial and temporal scales, this results in severe boundary discontinuity problems [1], [12], [14]. Hence, a more comprehensive approach, which integrates multiple sources of data and incorporates complex biogeochemical processes, may be necessary to accurately estimate global surface ocean DIC concentrations.

The distribution of DIC fields is strongly influenced by variations in SST and SSS, which account for approximately 70%–80% of the observed variability in DIC fields. Therefore, the changes in SST and SSS have a significant impact on the biogeochemical cycles of carbon and other elements in the ocean [1], [2], [6], [10]. The concentration of DIC in the ocean is also influenced by a range of biological processes (Chla), such as photosynthesis, respiration, and decomposition [1], [8], [14]. These processes can affect the balance of CO_2 in the water and thereby impact the global carbon cycle and ocean potential of hydrogen (pH) concentration. To accurately predict the future state of the ocean, it is necessary to comprehend the complex relationship between physical (SST and SSS) and biological (Chla) factors that influence DIC variability. Thus, the interaction between these factors must be taken into account when developing DIC models. Specifically, SST strongly influences DIC in the tropical regions [7], [11], while SSS plays a major role in the subtropical regions [4], [5], [11].

Sasse et al. [5] devised a self-organizing multiple linear output (SOMLO) strategy to forecast DIC levels in the mixed layer of the global oceans. This approach relied on physical (SST and SSS) and biochemical (nutrients and DO) parameters to produce accurate DIC predictions. In order to generate more precise DIC fields, Takahashi et al. [4] constructed global climatological DIC maps by considering in situ measurements from the past few decades after eliminating data from intra/interannual variability regions of the equatorial Pacific Ocean and global coastal regions. This approach aimed to provide a more consistent view of the global surface ocean DIC distribution and to identify the long-term trends and patterns. Recently, Broullón et al. [2] developed a feedforward neural network (FFNN) method for estimating spatial and depthwise variations of global monthly climatological DIC fields, which enables a more comprehensive understanding of global DIC patterns. Furthermore, Gregor and Gruber [15] created global surface ocean DIC maps based on the geospatial random cluster ensemble regression (GRaCER) method by utilizing data from the surface ocean CO_2 Atlas (SOCAT) to obtain the observations of partial pressure of CO_2 (pCO_2) and data from the GLODAPv2.2023 to obtain total alkalinity (TA). However, the estimation of DIC from chemical parameters, such as pCO_2 and TA, can be challenging due to a range of factors, including the sources of variability and error propagation, the presence of other DIC species, spatial and temporal variability, and external factors, such as anthropogenic CO_2 emissions and climate change [1]. More recently, Carroll et al. [16] investigated the space-time variability (over the period from 1995 to 2018) of DIC fields in the global ocean with the help of estimating the circulation and climate of the ocean (ECCO)—Darwin ocean biogeochemistry state model. In the polar regions, the dominant parameter affecting DIC is Chla [1], [9].

These algorithms have proved successful in improving surface ocean DIC estimations in global ocean regions, which involves the more complex multistage process to estimate the DIC fields. To overcome these challenges, the present study proposed a novel multiparametric regression (MPR) approach by considering the trends of DIC with respect to SST, SSS, and Chla concentration. The MPR model is formulated and validated using a large number of quality-controlled in situ measurements (SST, SSS, Chla, and DIC) collected over various coastal and open oceanic regions. The development of a parametric relationship for DIC with respect to physical (SST and SSS) and biological (Chla) parameters has enabled researchers to estimate the DIC concentration in different ocean regions using a multiparameter MLR approach. While the MLR approach has proven to be effective in estimating the distribution of DIC fields on a basin scale, it is not adequate for application to the entire global ocean region. This is due to the factors, such as the formulation and improper selection of model parameters, lack of in situ measurements, and the consideration of adequate global trends that are present in the different global oceanic regions. Krishna et al. [1] developed a multiparametric nonlinear regression (MPNR) approach that considered the complex global trends of pCO_2 in relation to physical and biogeochemical factors, such as SST, SSS, and Chla. This approach showed improved results over the global oceanic regions, motivating us to develop a similar regression-based DIC approach that could be applicable to the entire global oceans.

II. Data and Methods

A. In Situ Data

In order to develop and validate the MPR model, the in situ measurements (SST, SSS, Chla, and DIC) were obtained from the Global Ocean Data Analysis Project (GLODAPv2.2023) (https://www.glodap.info/index.php/data-access/), which provides a comprehensive collection of high-quality-controlled oceanographic data. Several research institutes have contributed in situ data to GLODAP by conducting multiple oceanographic cruises that covered various regions of the ocean, including the Atlantic, Pacific, Southern, Indian, and Arctic oceans. These cruises were conducted at different times

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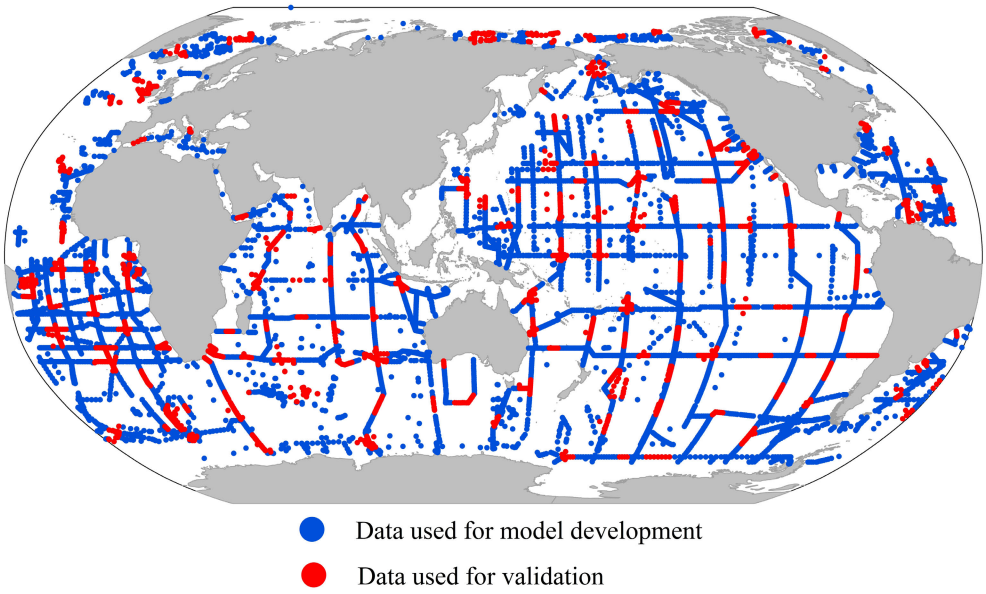


Fig. 1. Depicts a map of the in situ DIC sampling locations from the surface ocean that was utilized in this study.

throughout the year to capture the seasonal variability of DIC fields in the ocean. The data collected from these diverse sources provide a comprehensive and valuable resource for understanding the distribution and dynamics of DIC in the Global oceans. The in situ measurements encompass a wide range of spatial and temporal data, including seasonal and interannual variations, spanning the period from 2002 to 2021. Lauvset et al. [3] provided the detailed information regarding the cruise programs, measurement protocols, and associated uncertainties. For the development and validation of the MPR approach, we obtained a total of 12 800 in situ measurements. Out of these, 6000 data points were allocated for model development, while the remaining data points were reserved for model validation. This division of data points was determined by considering the spatiotemporal coverage and the availability of satellite data. Fig. 1 illustrates the spatial distribution of the in situ measurements. Although these measurements may not be evenly distributed across the ocean, they cover a diverse range of physical and biogeochemical processing regions, such as areas where physical mixing occurs, warming, stratified, major gyre systems, and the regions with high biological production. Therefore, the in situ data obtained from these measurements offer a unique and valuable perspective on the complex processes that occur in different parts of the ocean.

B. Satellite Data

To demonstrate the reliability and accuracy of the MPR approach in deriving spatiotemporal variability of global surface ocean DIC fields, we acquired global-mapped MODIS-Aqua sensor (spatial resolution: 4 × 4 km and temporal resolution: one day) Level-3 SST and Chla products from the National Aeronautics and Space Administration (NASA)-Goddard Space Flight Centre (GSFC) (https://oceancolor.gsfc.nasa.gov). In addition to this, global-mapped Level-4 SSS products from the Jet Propulsion Laboratory (JPL) (https://podaac.jpl.nasa.gov). The SST data can be considered accurate to ±4° [17], whereas the MMOI-SSS accuracy is around 0.19 PSU [18], and the uncertainty of Chla is about ±35% [19]. For the purpose of satellite data analysis, the NASA SeaWiFS Data Analysis System (SeaDAS) software was employed. To validate the satellite data, matchups were created by averaging the in situ measurements that corresponded to the satellite pixels at a spatial resolution of 0.25°. The spatiotemporal variations of surface ocean DIC fields were characterized using monthly, seasonal, and annual data (SST, SSS, and Chla) for the reference year 2017.

C. Theoretical Formulations of DIC

The ocean acts as a major sink for carbon, absorbing a significant portion of the CO_2 emitted into the atmosphere by anthropogenic activities [1], [4]. The absorbed CO_2 interacts with ocean water (H_2O) to form carbonic acid (H_2CO_3). This carbonic acid can subsequently dissociate into bicarbonate (HCO_3^-) and H^+ ions. Over longer timescales, a portion of the bicarbonate ions can further dissociate to form carbonate ions (CO_3^{2-}). These intricate alterations in carbonate chemistry can be comprehensively described as follows:

$$CO_{2(aq)} + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons HCO_3^- + H^+ \rightleftharpoons CO_3^{2-} + 2H^+. \tag{1}$$

The aforementioned series of chemical reactions ultimately leads to the formation of DIC, which includes inorganic forms of carbon, such as CO_2, HCO_3^-, and CO_3^{2-} ions

$$\mathrm{DIC} = \left[CO_3^{2-}\right] + \left[HCO_3^-\right] + [H_2CO_3] + \left[CO_{2(aq)}\right] \pm \text{minor constituents}. \tag{2}$$

Understanding the carbonate chemistry of DIC in the ocean is crucial for predicting how changes in these factors can affect

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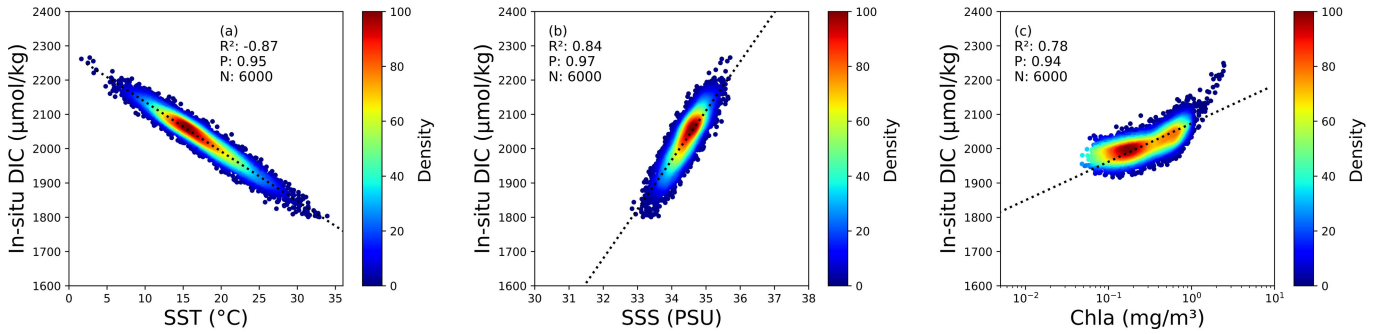


Fig. 2. Scatterplot shows the relationship between DIC and (a) SST, (b) SSS, and (c) Chla concentrations.

the concentration and balance of dissolved carbon species in seawater. For example, ocean acidification caused by increase CO_2 emissions can lower the pH of seawater, leading to a decrease in the concentration of carbonate ions. Hence, there is a need to estimate the spatiotemporal variability of DIC fields in the global oceans. Estimating the distribution of DIC in the ocean through chemical species can be challenging due to the complex and dynamic nature of seawater carbonate chemistry [1]. In addition, the accurate estimation of the spatiotemporal variability of equilibrium and carbonate dissociation constants (1) is difficult [1], which can make it challenging to quantify the different forms of DIC. This complexity is compounded by the fact that the carbonate system is influenced by a range of factors, including temperature, salinity, and biological processes, which can also vary over space and time [2], [4], [16]. Therefore, it is often more feasible to estimate DIC concentrations in the ocean using remote sensing or other techniques that do not rely on direct measurements of the seawater carbonate chemistry.

The advancements in satellite-based DIC estimates in recent times have revolutionized our ability to monitor DIC concentrations in the ocean with greater precision. These advances have yielded crucial insights into the carbon cycle of the ocean and its response to current climate change scenarios. Based on the greater advantages of satellite oceanographic data, multiple DIC studies were carried out using complex physical and biogeochemical parameters [2], [5], [8], [9], [10]. It should be reported that the spatiotemporal variability of surface ocean DIC is mainly controlled by the changes in SST, SSS, and Chla [1], [8], [9]. Therefore, a multiparametric MLR regression [8], [9] approach has been utilized to estimate the basin-scale distribution of DIC as a function-parametric regression approach that established a parametric relationship of these three parameters. Thus, in the present study, we formulated a multi between SST, SSS, Chla, and DIC as follows:

$$\mathrm{DIC} = \mathrm{ASST} + \mathrm{BSSS} + \mathrm{Clog(Chla)} + \mathrm{D}. \tag{3}$$

Phytoplankton biomass changes are intricately linked to multiple variables, including light availability, nutrient levels, and seawater properties [20]. The complex and dynamic nature of phytoplankton biomass, which undergoes diurnal variations in response to physical factors and light fluctuations, makes it challenging to establish a simple linear relationship between DIC and Chla concentration [1], [21]. To better account for the biological effects on DIC variations, we employed the logarithm of Chla ($\log_{10}$Chla) as a more suitable parameter. In order to improve the accuracy and precision of the regression coefficients in the MPR approach (3), we obtained in situ measurements from the diverse global oceanic regions. The objective of utilizing measurements from these regions was to improve our understanding of the intricate processes underlying DIC variability and produce more dependable outcomes. In addition to capture the temporal variability of DIC fields, the MPR model included a variety of seasonal and interannual in situ measurements. These measurements were integrated into the model to enable the characterization of fluctuations in DIC levels over time. Incorporating this temporal aspect allowed us to develop a more comprehensive comprehension of the various factors that impact DIC fields in the global oceans.

The MPR approach was developed based on a comprehensive set of 6000 in situ measurements. After a thorough analysis of these measurements, we have identified strong correlations of DIC with respect to SST, SSS, and Chla concentration. For instance, DIC exhibits a negative correlation with SST, indicating a decrease in DIC concentrations as SST increases. Conversely, a positive correlation is observed between DIC and SSS, as well as *Chla* concentration, signifying an increase in DIC concentrations with rising SSS and Chla levels. Fig. 2 illustrates the correlations between DIC and the influencing parameters, such as SST, SSS, and Chla. Based on these trends, we have formulated a single MPR equation that could be applied to the entire global oceans

$$\mathrm{DIC} = -9.082\mathrm{SST} + 36.114\mathrm{SSS} - 32.128\log(\mathrm{Chla}) + 936.05. \tag{4}$$

To overcome the challenges of rapidly changing oceanic conditions and accurately discern DIC trends, numerous studies have estimated DIC trends over physically and biologically defined regions using multiparameters derived from in situ or satellite observations. The MPR approach, in particular, is designed to estimate DIC levels as a function of all the influencing parameters that have a significant correlation with DIC. This approach allows for the estimates of DIC on large spatial and temporal scales, which may be difficult or expensive to measure directly. Additionally, to better understand the individual effects of each parameter (SST, SSS, and Chla)

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TABLE I
VARIOUS COMBINATIONS OF SINGLE AND MULTIPARAMETER DIC MODELS USED IN THE PRESENT STUDY

Single and multi-parameter DIC Models	
Model-1	DIC = -5.672SST+2135.5
Model-2	DIC = 21.787SSS+1319.9
Model-3	DIC = 80.438$\log_{10}$(Chla)+2015.5
Model-4	DIC = -5.645SST+1.397SSS+2088.5
Model-5	DIC = -4.997SST-4.997$\log_{10}$(Chla)+2125.6
Model-6	DIC = 0.261SSS+80.628$\log_{10}$(Chla)+2036.5
Model-7	DIC = -9.082SST+36.114SSS-32.128$\log_{10}$(Chla)+936.05

on DIC, both single and multiparameter DIC approaches have been formulated using the same model-developed data. This enabled us to understand the individual impact of each parameter on DIC levels and to determine which combinations provided the best fit to the in situ data. The regression coefficients of these models are presented in Table I. By examining the relationship between each parameter and DIC, we were able to optimize the models and reduce the residual errors between the model-derived and in situ DIC fields. This approach allowed us to develop more accurate DIC models that could account for the complex interactions between different parameters, ultimately improving our understanding of the factors that influence DIC variability in the global oceans. The implementation process is visually represented in Fig. 3.

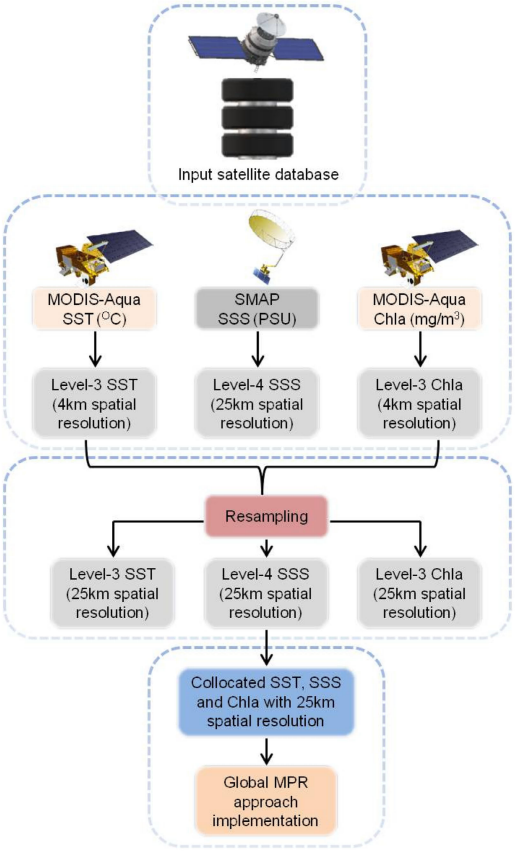


Fig. 3. Flowchart demonstrates the architecture of the MPR approach.

III. PERFORMANCE ASSESSMENT

To evaluate the effectiveness of the MPR regression approach in estimating DIC fields, a quantitative performance assessment is carried out using several standard statistical matrices. The mean relative error (MRE) matrix is used to calculate the systematic errors, while the root mean square (RMSE) matrix is used to calculate the random errors associated with the approach. The mean normalized bias (MNB) matrix is used to assess the deviation of the estimated DIC values from the measured in situ DIC values. In addition to these matrices, the slope, intercept, and correlation coefficient (R^2) are calculated through regression analysis between the estimated DIC values and the measured in situ DIC values. The slope and intercept provide information on the accuracy of the MPR approach in estimating the DIC values, while the R^2 indicates the degree of correlation between the estimated and measured in situ DIC values. By utilizing these statistical matrices, the performance of the MPR approach can be quantitatively evaluated and compared to other approaches. In particular, the MRE and RMSE matrices can provide information on the magnitude of the errors associated with the MPR approach, while the MNB, intercept, slope, and R^2 can provide insights into the direction and nature of these errors. Furthermore, the quantitative performance assessment of the MPR approach helps validate the accuracy and reliability of the approach in estimating DIC values and provides valuable information for improving and refining the approach in the future

$$\text{MRE} = \frac{1}{N}\sum_{i=1}^{N}\frac{|(\text{DIC}_{\text{estimated}} - \text{DIC}_{\text{in-situ}})|}{\text{DIC}_{\text{in-situ}}} \tag{5}$$

$$\text{MNB} = \frac{\sum_{i=1}^{N}(\text{DIC}_{\text{estimated}} - \text{DIC}_{\text{in-situ}})}{N} \tag{6}$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^{N}(\text{DIC}_{\text{estimated}} - \text{DIC}_{\text{in-situ}})^2}{N}}. \tag{7}$$

IV. RESULTS AND DISCUSSION

In this section, we present the validation of satellite-based single and multiparameter DIC models through intercomparison analysis. For the analysis, we utilized an independent in situ DIC dataset collected from open and coastal waters across the global ocean. Additionally, we describe the spatial and temporal variations (monthly/seasonal/interannual) of global surface ocean DIC fields using satellite data.

A. Evolution of Various Single and Multiparameter DIC Models Using In Situ and Satellite Observations

The MPR model is formulated by considering parametric relations of DIC with SST, SSS, and Chla, as these parameters have a significant impact on DIC variability in different regions of the ocean. The purpose of comparing different DIC models in the present study is to determine the most precise

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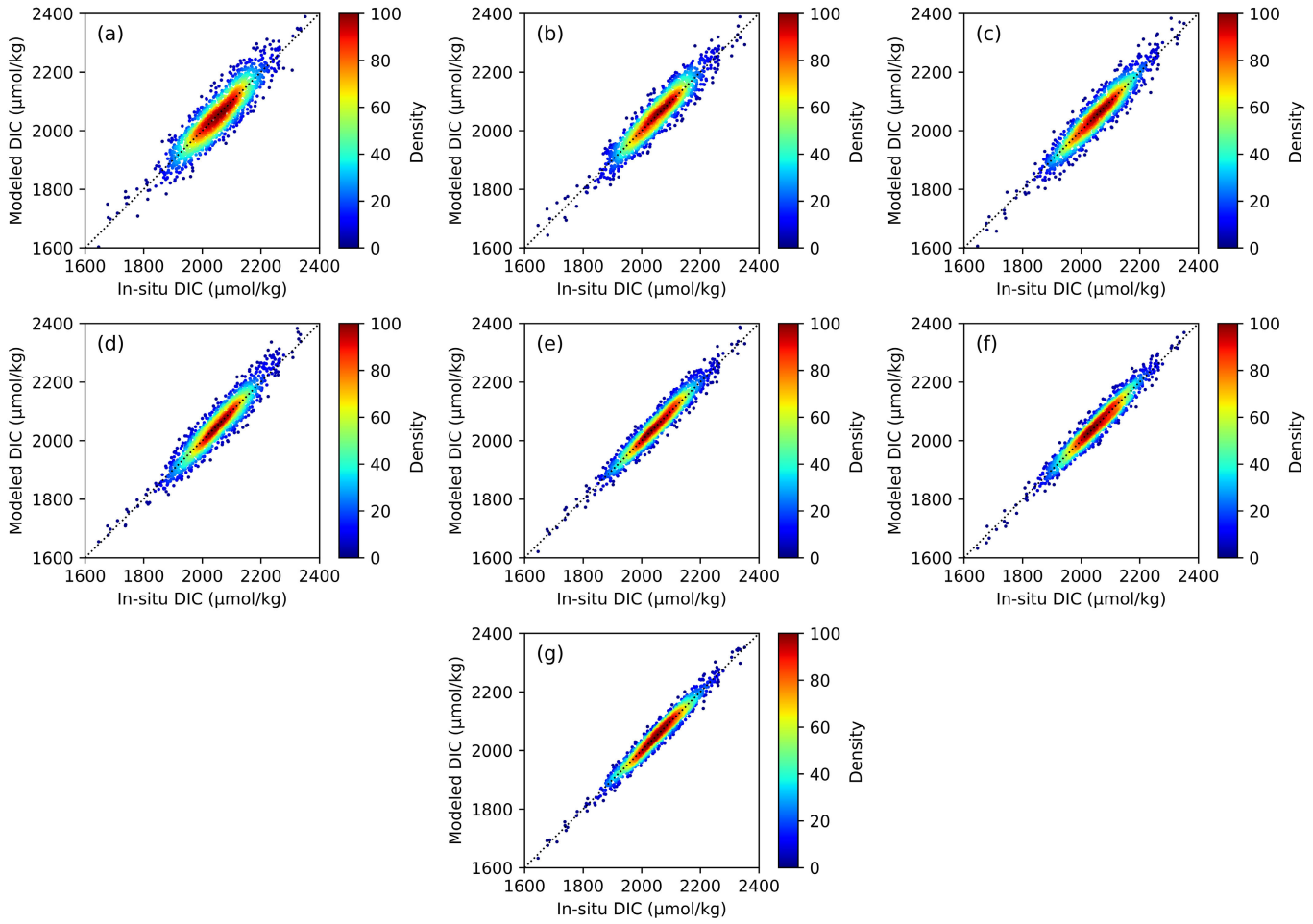


Fig. 4. Scatterplots (a)–(g) and shows the comparison of in situ DIC with in situ- and satellite-derived DIC values for the different DIC models (Models 1–7) (see Table II).

and dependable method for estimating DIC concentrations under varying oceanographic conditions. In order to assess the effectiveness of different DIC models, independent in situ measurements ($N = 6000$) were utilized for both in situ and satellite validation. However, due to cloud coverage, missing scan lines, and missing satellite product data, the number of matchups was reduced to a significant extent. In addition to this, the in situ measurements provide continuous data while satellite products are produced at a different spatial resolution (MODIS-Aqua: 4 km and SMAP: 0.25°), and it was necessary to generate matchup data by converting in situ observations into 0.25° spatial resolution satellite-derived products. The nearest neighborhood method was employed resample the MODIS-Aqua SST (4-km spatial resolution) and Chla (4-km spatial resolution) products to match the spatial resolution of the MMOI-SSS (0.25°) data due to its simplicity and computational efficiency [22]. This yielded in total of 2130 co-located DIC and corresponding SST, SSS, and Chla, which were used for in situ and satellite validation. Fig. 4 offers a detailed visual exploration of our study findings. It includes seven density scatterplots [labeled (a)–(g)]. These plots are designed to facilitate an in-depth comparison between direct in situ measured DIC values and the modeled DIC values derived from both in situ measurements and satellite-derived products. This comparative analysis encompasses seven distinct DIC models (Models 1–7), each providing a unique perspective on the DIC estimations. This allows for a more robust and holistic assessment of the different DIC model performances across a diverse range of data sources, ultimately enhancing the comprehensiveness and reliability of our findings.

Table II offers a comprehensive and detailed statistical comparison of Models 1–7. The validation results indicate that the MPR-7 model for DIC fields demonstrates a remarkable agreement with both in situ-derived data (MRE 0.008, MNB 0.013, RMSE 5.43, and R^2 0.96) and satellite-derived data (MRE 0.078, MNB 0.012, RMSE 8.21, and R^2 0.92), signifying a notably low level of error. In contrast, the single and multiparameter DIC models (Models 1–6) displayed comparatively higher statistical values when compared to both in situ-derived (MRE 0.011–0.031, MNB 0.025–0.321, RMSE 7.32–7.84, and R^2 0.82–0.92) and satellite-derived (MRE 0.075–0.092, MNB −0.178–0.362, RMSE 8.83–11.54, and R^2 0.81–0.89) DIC fields. As a result, Model 7 was selected as the final MPR regression equation and subsequently applied to the satellite data for implementation.

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TABLE II
STATISTICAL INTERCOMPARISON OF VARIOUS COMBINATIONS OF DIC MODELS

MODEL	Validation	MRE(μmol/kg)	MNB(μmol/kg)	RMSE(μmol/kg)	R^2
Model-1	In-situ	0.019	0.031	7.43	0.82
	Satellite	0.092	-0.178	11.24	0.81
Model-2	In-situ	0.031	0.035	7.32	0.86
	Satellite	0.078	-0.125	11.54	0.84
Model-3	In-situ	0.025	0.025	7.74	0.88
	Satellite	0.092	0.045	10.87	0.82
Model-4	In-situ	0.011	0.018	7.84	0.86
	Satellite	0.075	0.015	9.86	0.83
Model-5	In-situ	0.016	0.321	7.74	0.88
	Satellite	0.083	0.362	8.83	0.84
Model-6	In-situ	0.014	0.051	7.54	0.92
	Satellite	0.078	-0.058	8.85	0.89
Model-7	In-situ	0.008	0.013	5.43	0.96
	Satellite	0.078	0.012	8.21	0.92

TABLE III
STATISTICAL PERFORMANCE OF MPR APPROACH IN VARIOUS OCEANIC REGIONS

Region	Validation	MRE(μmol/kg)	MNB(μmol/kg)	RMSE(μmol/kg)	R^2
Arctic	In-situ	0.004	0.006	5.62	0.93
	Satellite	0.141	0.014	7.96	0.91
Atlantic	In-situ	0.003	0.004	5.36	0.97
	Satellite	0.142	0.015	7.68	0.92
Pacific	In-situ	0.007	0.011	6.72	0.94
	Satellite	0.064	0.013	8.54	0.92
Indian	In-situ	0.012	0.011	5.38	0.96
	Satellite	0.094	0.055	7.92	0.93

B. Evolution of MPR Performance Over Various Regional Ocean Waters

In order to assess the performance of the MPR approach in diverse regional oceanic waters, we applied the MPR methodology to major oceanic regions. To accomplish this, the existing dataset of 2130 independent in situ measurements was strategically divided based on regional criteria. The MPR approach was rigorously applied to each regional specific data, and the results were compared to both in situ- and satellite-derived data (see Fig. 8). The comparative analyses and corresponding line plot were presented in Fig. 7, whereas corresponding statistics are presented in the Table V. The statistical comparisons revealed that the MPR method works very well in all these different ocean regions. However, there are slight deviations/uncertainties in all the ocean regions caused due to the influence of other parameters. The spatiotemporal variability of DIC in the ocean is subject to a multitude of factors that extend beyond the capabilities of satellite observations. These factors include complex processes, such as microbial respiration, remineralization, and grazing, which are challenging to estimate through remote sensing approach. In addition to this, the spatiotemporal variability of surface ocean DIC is influenced by other contributing parameters. These parameters include DO, AOU, and nutrients concentration (nitrogen, phosphorous, and Si concentration). Unfortunately, we have not incorporated these additional parameters into our study due to the unavailability of accurate and precise satellite-based models capable of estimating nutrients, DO, and AOU in the vast expanse of the global ocean. The primary goal of the present MPR approach is to generate global surface ocean DIC maps on a large-scale using satellite-derived oceanographic data. For this purpose, we rely on major dominant parameters readily accessible from satellite observations. These parameters serve as reliable proxies for DIC levels. However, to ensure the reliability of this estimation and to quantify the uncertainty introduced by these key parameters (SST, SSS, and Chla), we conducted a thorough satellite validation. Tables III and IV present the performance metrics of the DIC model for various oceanic regions and a regional breakdown

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TABLE IV
REGIONAL SATELLITE VALIDATION STATISTICS OF SST, SSS, AND CHLA

Region	Data	Validation	MRE (μmol/kg)	MNB (μmol/kg)	RMSE (μmol/kg)	R^2
Arctic	SST	Satellite	0.041	0.045	7.24	0.94
	SSS	Satellite	0.088	0.012	6.96	0.91
	Chla	Satellite	0.018	0.063	7.12	0.93
Atlantic	SST	Satellite	0.012	0.014	7.29	0.92
	SSS	Satellite	0.082	0.026	6.89	0.95
	Chla	Satellite	0.012	-0.015	7.21	0.93
Pacific	SST	Satellite	0.081	0.321	6.56	0.93
	SSS	Satellite	0.074	0.051	7.31	0.91
	Chla	Satellite	0.011	0.015	6.98	0.94
Indian	SST	Satellite	0.036	0.053	7.23	0.92
	SSS	Satellite	0.014	-0.037	6.84	0.93
	Chla	Satellite	0.075	0.024	7.41	0.91

of our satellite-based comparisons for SST, SSS, and Chla concentration.

These comparisons are crucial in evaluating the consistency and accuracy of the satellite-derived data when compared to in situ measurements. The validation results revealed that the uncertainties associated with these critical parameters remain within the predetermined range suitable for satellite remote sensing methodologies. This reinforces the robustness of our approach and the trustworthiness of the data we employ for estimating DIC in oceanic regions.

C. Model Sensitivity Analysis

The DIC model (Model 7) performance was evaluated in each scenario by calculating key performance metrics, including RMSE, MRE, MNB, and R^2 (see Fig. 5). When SST increased by +1 °C, the DIC model showed an RMSE of 8.01 μmol/kg, an MRE of 0.018 μmol/kg, an MNB of 0.023 μmol/kg, and an R^2 of 0.96, indicating a relatively high level of accuracy in estimating DIC values under increased temperature conditions. Conversely, when SST decreased by −1 °C, the model exhibited slightly lower error metrics with an RMSE of 8.67 μmol/kg, an MRE of 0.092 μmol/kg, an MNB of 0.124 μmol/kg, and an R^2 of 0.94, suggesting improved performance under decreased temperature conditions. Variations in SSS also influenced the DIC model performance. When SSS increased by +1, the model showed an RMSE of 8.94 μmol/kg, an MRE of 0.096 μmol/kg, an MNB of 0.089 μmol/kg, and an R^2 of 0.93, indicating decreased estimation accuracy. Conversely, when SSS decreased by −1, the model exhibited higher error metrics with an RMSE of 8.46 μmol/kg, an MRE of 0.125 μmol/kg, an MNB of 0.092 μmol/kg, and an R^2 of 0.92, suggesting reduced performance under decreased salinity conditions. Moreover, variations in Chla concentrations had an impact on the DIC model performance. When Chla was increased by +20%, the model exhibited an RMSE of 9.18 μmol/kg, an MRE of 0.084 μmol/kg, an MNB of 0.045 μmol/kg, and an R^2 of 0.95, indicating decreasing accuracy in predicting DIC values under elevated chlorophyll concentrations. Conversely, when Chla was decreased by −20%, the model showed slightly higher error metrics with an RMSE of 9.24 μmol/kg, an MRE of 0.092 μmol/kg, an MNB of 0.087 μmol/kg, and an R^2 of 0.93 (see Table VI).

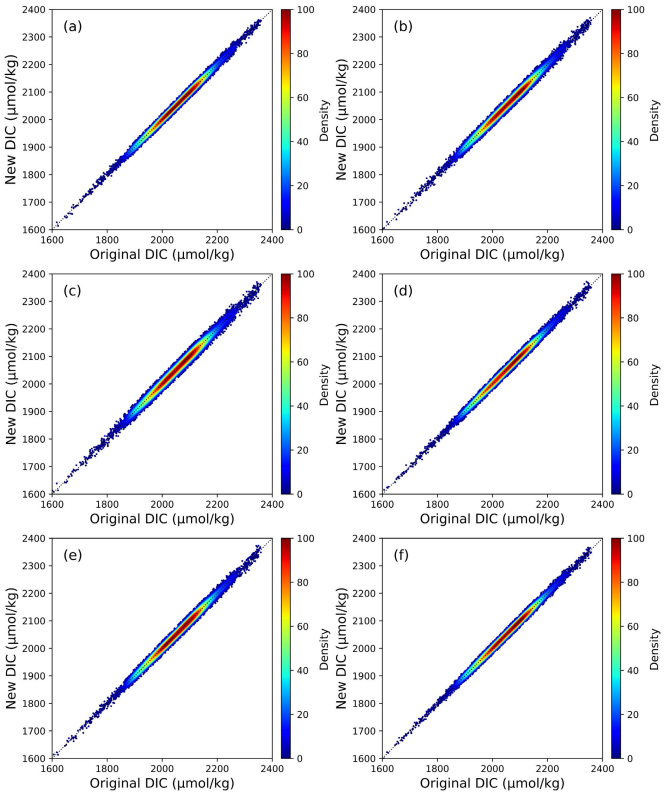


Fig. 5. DIC model sensitivity to change in the input (a) +1 °C SST, (b) −1 °C SST, (c) +1 SSS, (d) −1 SSS, (e) +20% Chla, and (f) −20% Chla based on the data used for model development.

D. Intercomparison of Present MPR Approach With the Existing Global DIC Studies

The spatiotemporal variability of DIC fields in the global surface ocean has been a subject of investigation through

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TABLE V
STATISTICAL PERFORMANCE OF GLOBAL DIC FIELDS

Region	Validation	MRE (μmol/kg)	MNB (μmol/kg)	RMSE (μmol/kg)	R^2	P	N
Global	Model derived DIC	0.008	0.013	5.43	0.98	0.97	1400
	Satellite derived DIC	0.078	0.012	8.21	0.96	0.94	1400

TABLE VI
STATISTICS OF THE SENSITIVITY OF THE DIC MODEL (MODEL 7) TO UNCERTAINTIES IN EACH OF THE PARAMETER SST, SSS, AND CHLA

Cases	**MRE** (μmol/kg)	**MNB** (μmol/kg)	**RMSE** (μmol/kg)	**R^2**
+1°C in SST	0.018	0.023	8.01	0.96
-1°C in SST	0.092	0.124	8.67	0.94
+1 in SSS	0.096	0.089	8.94	0.93
-1 in SSS	0.125	0.092	8.46	0.92
+20% in Chla	0.084	0.045	9.18	0.95
-20% in Chla	0.092	0.087	9.24	0.93

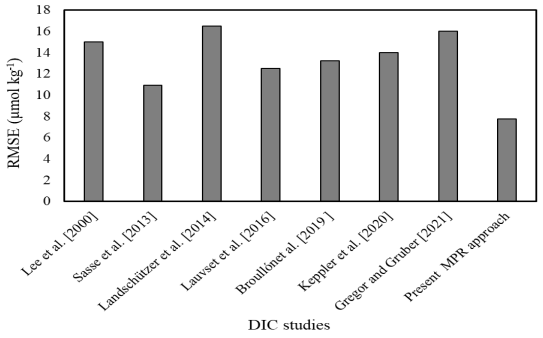


Fig. 6. Bar chart illustrating the contrast in RMSE between the present study and global DIC studies.

various in situ-based DIC approaches. These approaches have harnessed parameters, such as SST, SSS, Chla, Latitude, DO, AOU, and nutrient concentrations, to analyze DIC fluctuations. However, the application of these approaches to satellite data faces limitations, mainly due to the absence of precise and accurate satellite-based models capable of estimating certain parameters across global ocean regions. In the present MPR approach, we have considered only the dominant parameters, excluding the contribution from minor variables, such as DO, AOU, and nutrient concentrations. Recognizing the methodological disparities and parameter considerations in the present MPR approach and existing DIC studies, our present study has opted to employ RMSE deviations mentioned in the literature for the intercomparison purposes. This approach enables us to effectively compare and interconnect our MPR outcomes with those of previous DIC investigations (see Fig. 6).

E. Spatiotemporal Variability of Global DIC Fields

Understanding the magnitude and spatiotemporal variability of DIC fields is crucial for accurately modeling the carbon cycle of the ocean and predicting its response to changing environmental conditions, such as ocean acidification and global warming. Therefore, the MPR approach was applied to satellite data (SST, Chla, and SSS), enabling the generation of global maps depicting the spatiotemporal variability of surface ocean DIC fields (see Fig. 9). The variability of individual parameters, such as SST, SSS, and Chla, has a significant influence on the levels of DIC fields in the ocean. To better understand how these individual parameters affect the overall variability of DIC fields, their spatiotemporal variability is analyzed. This analysis provides insights into the contribution of each parameter to DIC variability and can enhance our understanding of the spatiotemporal variability of DIC fields in the global oceans.

The Atlantic Ocean is unique in terms of its DIC distribution, which varies significantly from the subtropical to subpolar regions. The DIC concentrations are generally higher in the subtropical regions, ranging from 1700 to 2000 μmol/kg, and lower in the equatorial and subpolar regions (see Fig. 9), ranging from 2000 to 2230 μmol/kg. The latitudinal gradient of DIC is mainly driven by the complex ocean circulation patterns in the North Atlantic and the relative homogeneity of DIC in the South Atlantic. The Gulf Stream system, for example, dominates the circulation in the North Atlantic and transports DIC-rich waters from the subtropical region to the subpolar region, leading to higher DIC concentrations in the North Atlantic [16]. However, the South Atlantic Ocean displays a uniform DIC distribution, primarily shaped by the mixing of various water masses, with concentrations ranging approximately between 1700 and 1950 μmol/kg. In the seasonal images (see Fig. 10), we typically observe higher DIC fields in the subtropical regions of the Pacific Ocean. This is primarily attributed to the elevated biological productivity in these areas. In contrast, the equatorial and subpolar regions tend to exhibit lower DIC concentrations. In the Indian Ocean, we observe that during the spring and winter seasons, DIC fields are generally lower compared to the summer and autumn seasons (see Fig. 10). This variation is primarily influenced by the circulation patterns driven by the monsoon. Notably, the DIC concentrations are relatively

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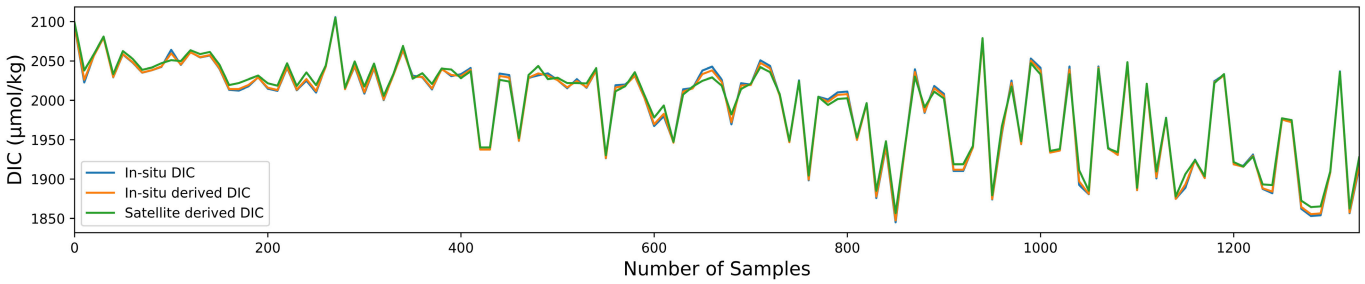


Fig. 7. Line plots illustrating the MPR approach performance over global ocean waters (see Table V).

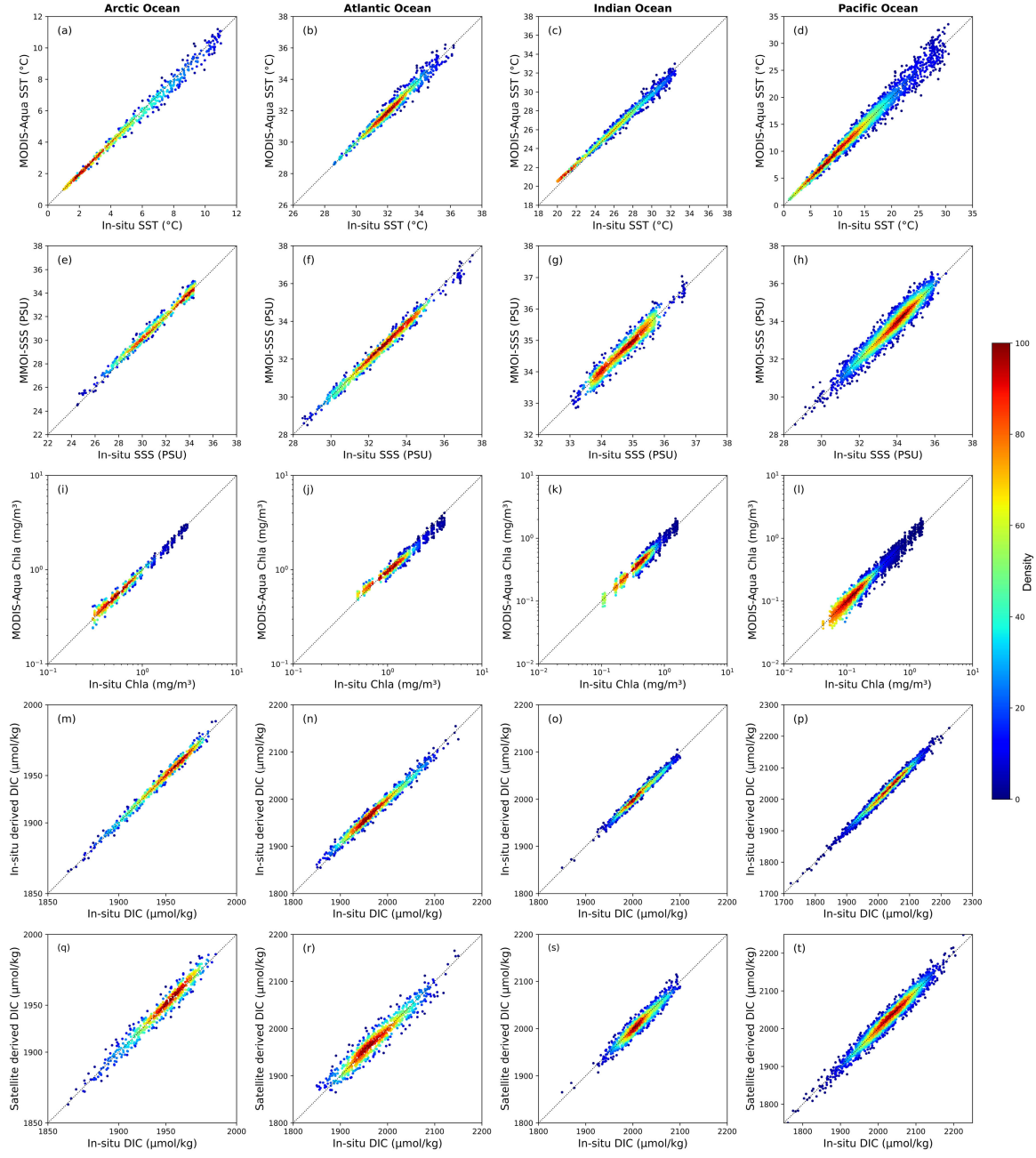


Fig. 8. Scatterplots show the comparison of (a)–(d) in situ- and satellite-derived SST, (e)–(h) in situ- and satellite-derived SSS, (i)–(l) in situ- and satellite-derived Chla, (m)–(p) direct in situ-measured and in situ-derived DIC, and (q)–(t) in situ- and satellite-derived DIC for the Atlantic, Arctic, Pacific, Indian, and Pacific Ocean regions (see Tables III and IV).

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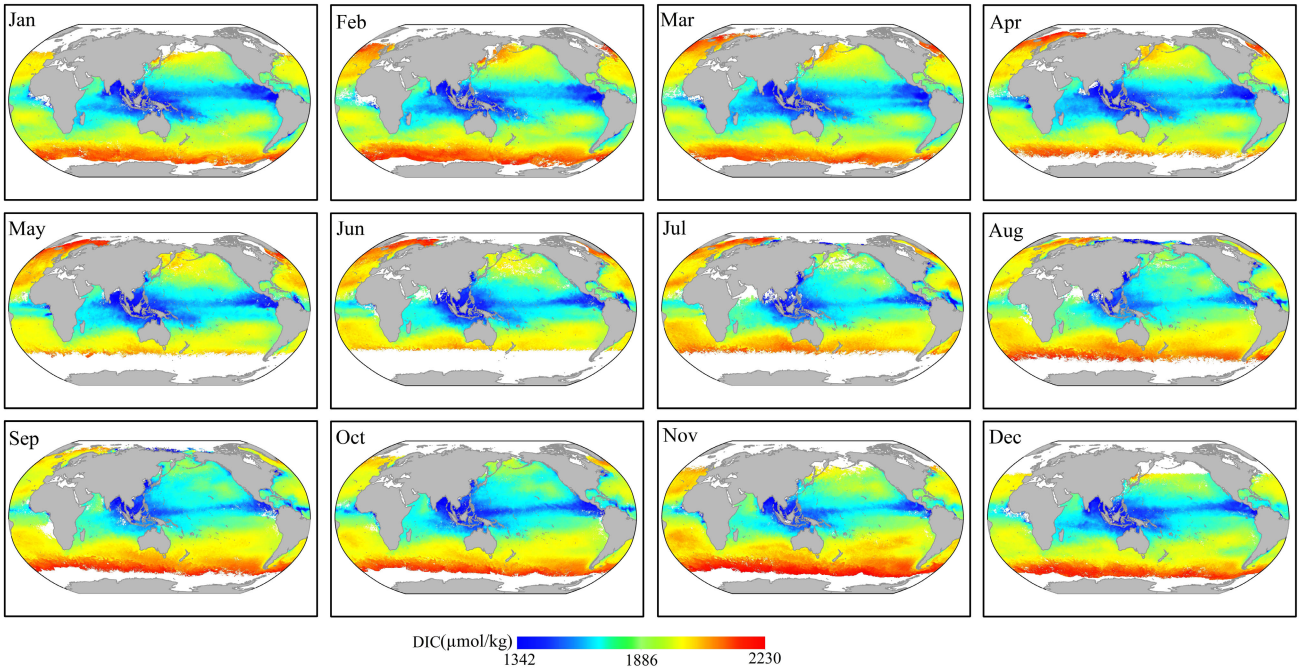


Fig. 9. Monthly global distribution of surface ocean DIC (μmol/kg) at 0.25° spatial resolutions for the reference year 2017.

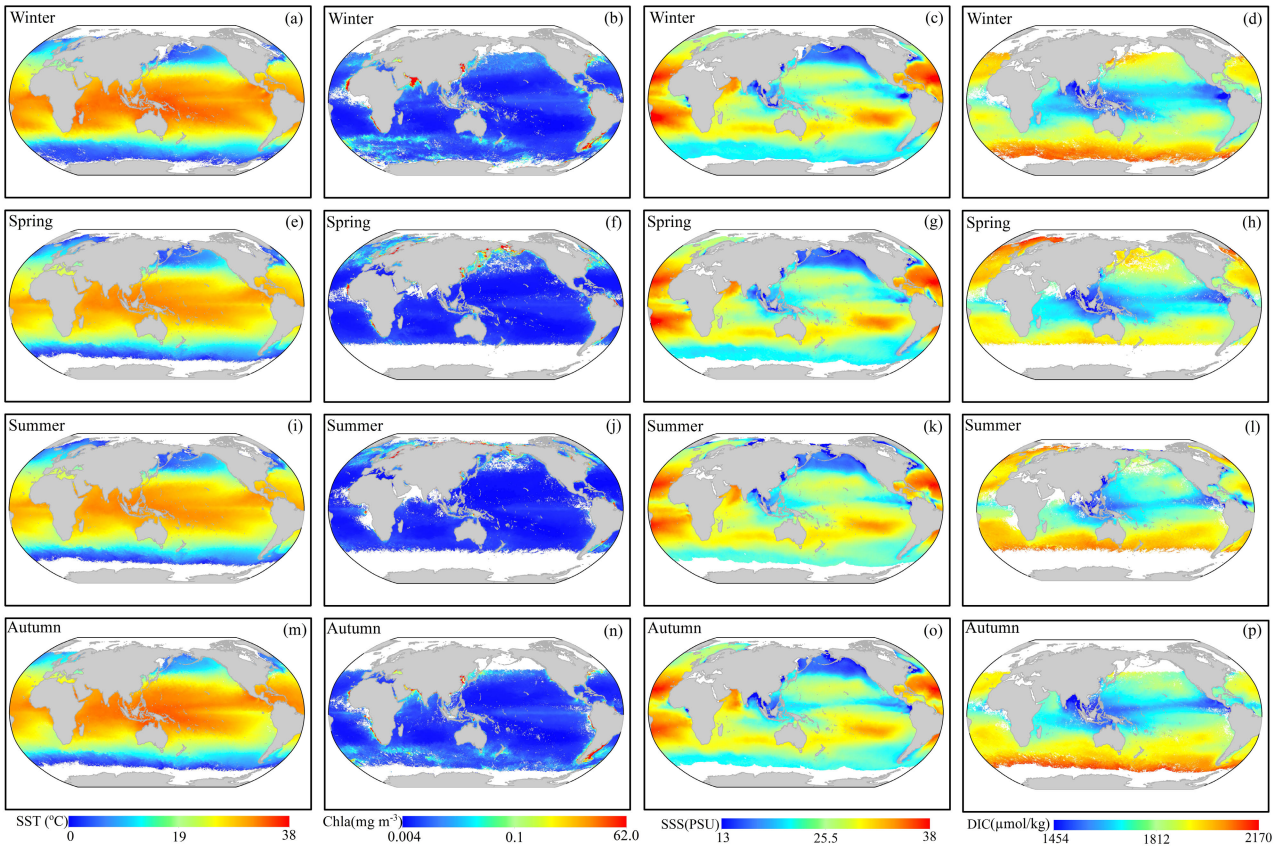


Fig. 10. Global seasonal distribution of SST, Chla, SSS, and DIC fields for the Winter [January–March, (a)–(d)], Spring [April–June, (e)–(h)], Summer [July–September, (i)–(l)], and Autumn [October–December, (m)–(p)] period for the reference year 2017.

higher in the Arabian Sea. This is attributed to the upwelling of nutrient-rich water in this region, which promotes heightened biological productivity. Consequently, marine organisms in the Arabian Sea absorb more DIC, contributing to the elevated DIC concentrations [8]. In contrast, the Bay of Bengal experiences lower DIC concentrations due to the influx of freshwater

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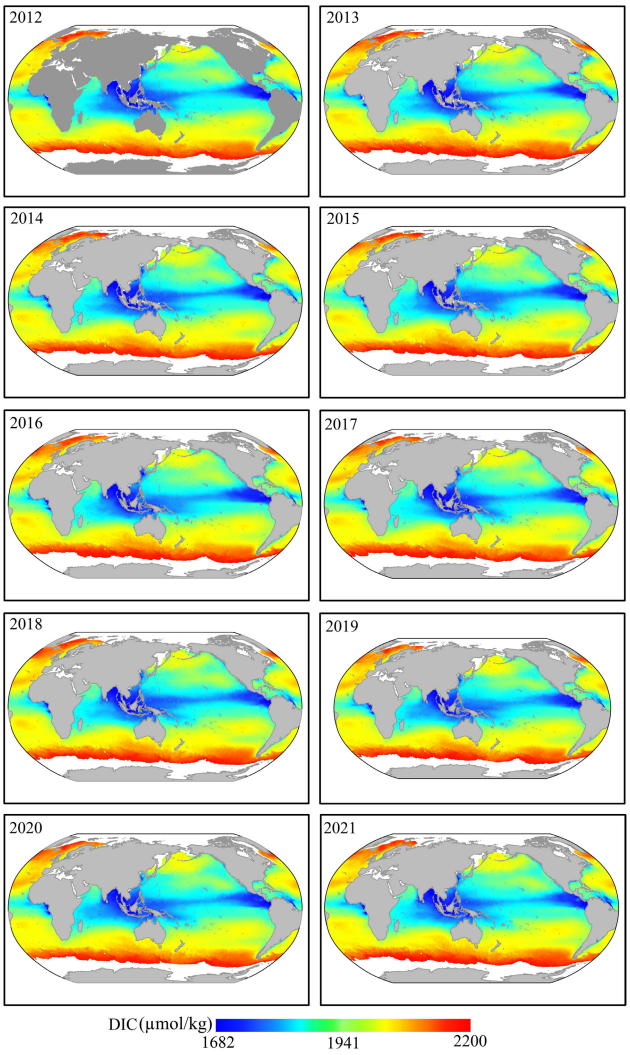


Fig. 11. Interannual variability of surface ocean DIC fields for the period 2012–2021.

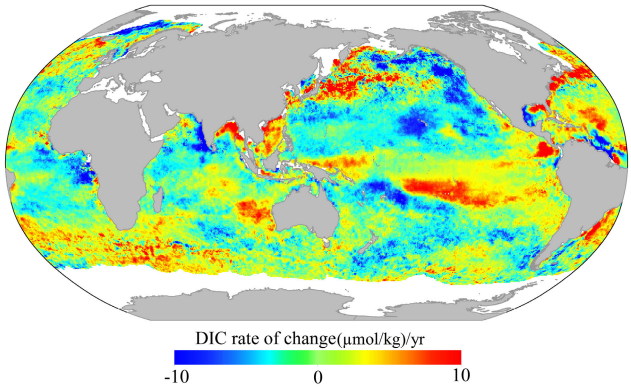


Fig. 12. Rate of change of DIC fields for the period 2012–2021.

from rivers, which leads to reduced biological productivity and lower DIC uptake by marine organisms [23]. The seasonal (see Fig. 10), rate of change (ROC) of DIC fields (see Fig. 12), and interannual variability (see Fig. 11) and for surface ocean DIC fields presented in this section play an important role for predicting future changes in the carbon cycle and its impacts on climate and marine ecosystems. The seasonal variability of surface ocean DIC is influenced by a range of factors, including temperature, biology, and mixing [2], [5], [16]. The ocean surface DIC concentrations are typically highest in the winter and lowest in the summer in the Northern Hemisphere and vice versa in the Southern Hemisphere (see Fig. 10). During the winter, colder temperatures lead to increased DIC concentrations, while in summer, warmer temperatures lead to decreased DIC concentrations [2], [5], [16], [24]. The annual DIC was computed as the mean of monthly DIC fields for each corresponding year. Annual DIC maps were generated for the period 2012–2021 to investigate interannual variability in the global ocean and to compute the DIC ROC in the global ocean. The ROC was computed using a linear least squares regression (LLSR) method. The LLSR was applied at each data point over a span of ten years. Thus, at each grid point, we have a list of ten DIC data points. A regression is performed on these ten points to determine the corresponding increasing or decreasing trend (see Fig. 12). The interannual variability of surface ocean DIC is influenced by a variety of large-scale oceanic and atmospheric processes. In a global context, the year-to-year changes are mainly influenced by vertical advection in equatorial areas, where ENSO is responsible for the most significant alterations in upper-ocean DIC mass (see Fig. 11) [16].

The North Atlantic Oscillation (NAO) also influences surface ocean DIC concentrations. During the positive phases of the NAO, the North Atlantic experiences enhanced mixing, leading to increased uptake of CO_2 and higher DIC concentrations [2], [5], [25]. The spatiotemporal structures of DIC fields observed in the global oceans are consistent with previous studies [2], [4], [5], [16]. This consistency reinforces the importance of continued research and monitoring of oceanic DIC dynamics for accurate modeling and prediction of changes.

V. Conclusion

A novel MPR approach was developed for estimating the spatiotemporal variability of global surface ocean DIC fields using physical (SST and SSS) and biological (Chla) parameters. Unlike previous models that relied on limited in situ data and multiple regression equations for particular regions/seasons and few confirmed DIC trends, the MPR approach demonstrated robustness in providing continuous coverage of DIC fields with greater accuracy by considering the global trends of DIC with respect to SST, SSS, and Chla concentration. The validation results using independent in situ data revealed that the DIC fields derived from the present MPR approach are very closely related to satellite-derived DIC fields. Despite the existence of multiple in situ and satellite-based approaches that consider complex physical and biogeochemical parameters to estimate the spatiotemporal variability of surface ocean DIC fields, the MPR approach is more accurate and reliable in estimating DIC from satellite-derived SST, SSS, and Chla products for the global region. Finally, the spatial (global) and temporal (monthly/seasonal/annual) variability of surface ocean DIC

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fields was demonstrated by considering the satellite oceanographic data (SST, SSS, and Chla) with a spatial resolution of 0.25° × 0.25°. The MPR approach was rigorously validated using both regional and global in situ data, indicating its potential as a promising tool for estimating global surface ocean DIC fields. This advancement could significantly enhance our current understanding of the global carbon cycle, air-sea CO_2 fluxes, ocean acidification, and changes in the biogeochemistry of the ocean, with an emphasis on the current climate change scenarios. In addition, this study suggests that the global surface ocean pCO_2, pH, and other carbonate chemistry parameters can be estimated from the inorganic carbonate chemistry calculations with the input of TA and DIC data. Although the MPR-based DIC estimates have a considerably small error, they may still be affected by various complex physical and biogeochemical processes that occur in certain oceanic regions, such as upwelling, organic matter formation and destruction, and cold environments in the polar oceans. Hence, our future research must focus on improving the accuracy of spatiotemporal estimates of ocean surface DIC fields by utilizing regional-specific in situ data collected from various oceanic regimes and including other relevant parameters.

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REFERENCES

[1] K. V. Krishna, P. Shanmugam, and P. V. Nagamani, "A multiparametric nonlinear regression approach for the estimation of global surface ocean pCO_2 using satellite oceanographic data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 6220–6235, 2020.

[2] D. Broullón et al., "A global monthly climatology of oceanic total dissolved inorganic carbon: A neural network approach," *Earth Syst. Sci. Data*, vol. 12, no. 3, pp. 1725–1743, Aug. 2020.

[3] S. K. Lauvset et al., "GLODAPv2.2022: The latest version of the global interior ocean biogeochemical data product," *Earth Syst. Sci. Data*, vol. 14, no. 12, pp. 5543–5572, Dec. 2022.

[4] T. Takahashi et al., "Climatological distributions of pH, pCO_2, total CO_2, alkalinity, and $CaCO_3$ saturation in the global surface ocean, and temporal changes at selected locations," *Mar. Chem.*, vol. 164, pp. 95–125, Aug. 2014.

[5] T. P. Sasse, B. I. McNeil, and G. Abramowitz, "A novel method for diagnosing seasonal to inter-annual surface ocean carbon dynamics from bottle data using neural networks," *Biogeosciences*, vol. 10, no. 6, pp. 4319–4340, Jun. 2013.

[6] K. R. Arrigo, S. Pabi, G. L. van Dijken, and W. Maslowski, "Air-sea flux of CO_2 in the Arctic ocean, 1998–2003," *J. Geophys. Res., Biogeosci.*, vol. 115, no. G4, pp. 1998–2003, Nov. 2010.

[7] N. R. Bates, A. C. Pequignet, and C. L. Sabine, "Ocean carbon cycling in the Indian ocean: 1. Spatiotemporal variability of inorganic carbon and air-sea CO_2 gas exchange," *Global Biogeochem. Cycles*, vol. 20, no. 3, pp. 1–13, Sep. 2006, Art. no. GB3020, doi: 10.1029/2005GB002491.

[8] V. V. S. S. Sarma, "Monthly variability in surface pCO_2 and net air-sea CO_2 flux in the Arabian sea," *J. Geophys. Res., Oceans*, vol. 108, no. C8, p. 3255, Aug. 2003.

[9] V. V. S. S. Sarma et al., "Basin-scale pCO_2 distribution using satellite sea surface temperature, chl a, and climatological salinity in the north Pacific in spring and summer," *Global Biogeochem. Cycles*, vol. 20, no. 3, pp. 1–13, Sep. 2006.

[10] R. N. Bates, A. C. Pequignet, and C. L. Sabine, "Ocean carbon cycling in the Indian ocean: 1. Spatiotemporal variability of inorganic carbon and air-sea CO_2 gas exchange," *Global Biogeochemical Cycles*, vol. 20, no. 3, 2006, doi: 10.1029/2005GB00249.

[11] K. Lee, R. Wanninkhof, R. A. Feely, F. J. Millero, and T. Peng, "Global relationships of total inorganic carbon with temperature and nitrate in surface seawater," *Global Biogeochem. Cycles*, vol. 14, no. 3, pp. 979–994, Sep. 2000.

[12] K. V. Krishna and P. Shanmugam, "Robust estimates of the total alkalinity from satellite oceanographic data in the global ocean," *IEEE Access*, vol. 11, pp. 42824–42838, 2023.

[13] D. W. R. Wallace, "Monitoring global ocean carbon inventories," *OOSDP Backgr. Rep. Number*, vol. 5, p. 54, Jan. 1995.

[14] W. T. Liu and X. Xie, "Space observation of carbon dioxide partial pressure at ocean surface," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 10, no. 12, pp. 5472–5484, Dec. 2017.

[15] L. Gregor and N. Gruber, "OceanSODA-ETHZ: A global gridded data set of the surface ocean carbonate system for seasonal to decadal studies of ocean acidification," *Earth Syst. Sci. Data*, vol. 13, no. 2, pp. 777–808, Mar. 2021.

[16] D. Carroll et al., "Attribution of space-time variability in global-ocean dissolved inorganic carbon," *Global Biogeochemical Cycles*, vol. 36, no. 3, Mar. 2022, Art. no. e2021GB007162.

[17] O. B. Brown et al., *MODIS Infrared Sea Surface Temperature Algorithm Algorithm Theoretical Basis Document Version 2.0*. Coral Gables, FL, USA: Univ. Miami, 1999, p. 98.

[18] O. Melnichenko et al., "Aquarius/SMAP sea surface salinity optimum interpolation analysis," IPRC, Kalanthapanai, India, Tech. Note No 7, 2021.

[19] T. S. Moore, J. W. Campbell, and M. D. Dowell, "A class-based approach to characterizing and mapping the uncertainty of the MODIS ocean chlorophyll product," *Remote Sens. Environ.*, vol. 113, no. 11, pp. 2424–2430, Nov. 2009.

[20] D. A. Siegel et al., "Regional to global assessments of phytoplankton dynamics from the SeaWiFS mission," *Remote Sens. Environ.*, vol. 135, pp. 77–91, Aug. 2013.

[21] B. Hales et al., "Satellite-based prediction of pCO_2 in coastal waters of the eastern north Pacific," *Prog. Oceanogr.*, vol. 103, pp. 1–15, Sep. 2012.

[22] M. B. Lyons et al., "A comparison of resampling methods for remote sensing classification and accuracy assessment," *Remote Sens. Environ.*, vol. 208, pp. 145–153, Apr. 2018.

[23] V. V. S. S. Sarma et al., "Sources and sinks of CO_2 in the west coast of Bay of Bengal," *Tellus B, Chem. Phys. Meteorol.*, no. 64, no. 1, Jan 2012, Art. no. 10961, doi: 10.3402/tellusb.v64i0.10961.

[24] L.-Q. Jiang, B. R. Carter, R. A. Feely, S. K. Lauvset, and A. Olsen, "Surface ocean pH and buffer capacity: Past, present and future," *Sci. Rep.*, vol. 9, no. 1, Dec. 2019, Art. no. 18624.

[25] S. S. Butcher, R. J. Charlson, G. H. Orians, and G. V. Wolfe, "Global biogeochemical cycles," in *Global Biogeochemical Cycles*, 1992, pp. 597–609.



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